



CLOUD TECHNOLOGIES

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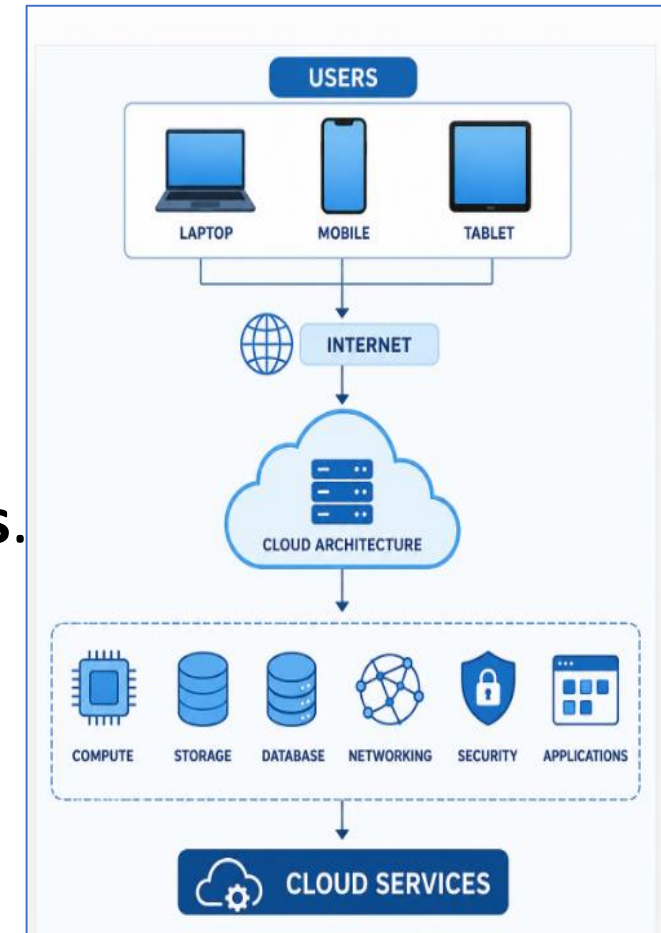
Cloud Architecture Design Principles

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At the end of this session, students will be able to:

- Define cloud architecture.
- Understand why architecture principles are important.
- Explain architecture design principles.
- Identify the characteristics of a good cloud architecture.
- Relate architecture principles to real-world cloud applications.

- A cloud application must serve millions of users simultaneously, requiring strong design principles.
- Cloud architecture unifies **hardware, software, virtualization, networking, storage, security & services.**
- Well-designed systems remain **scalable, reliable, secure, fault-tolerant, and cost-effective.**
- They ensure availability even during failures, cyberattacks, or sudden demand spikes. to the final service.



- **Key Points Why Architecture Principles Matter, without proper architecture:**
 - Systems become slow. , Downtime increases.
 - Maintenance becomes difficult.
 - Security risks increase.
 - Costs become higher.
 - Good architecture helps organizations provide uninterrupted services while minimizing operational costs.

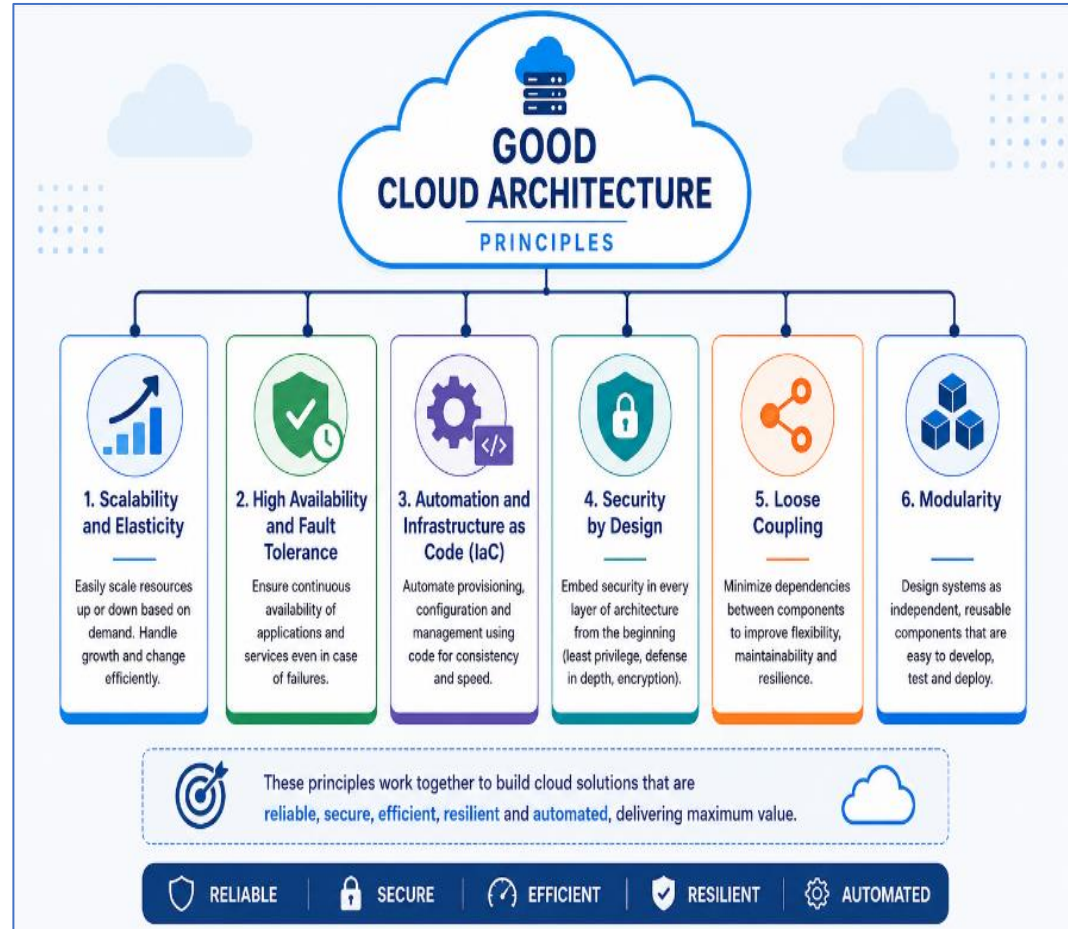
List five applications used every day that depend on cloud architecture

- Cloud architects follow several design principles to develop efficient cloud systems.
- **Major Design Principles**
 - Scalability and Elasticity
 - High Availability and Fault Tolerance
 - Automation and Infrastructure as Code (IaC)
 - Security by Design , Loose Coupling, Modularity

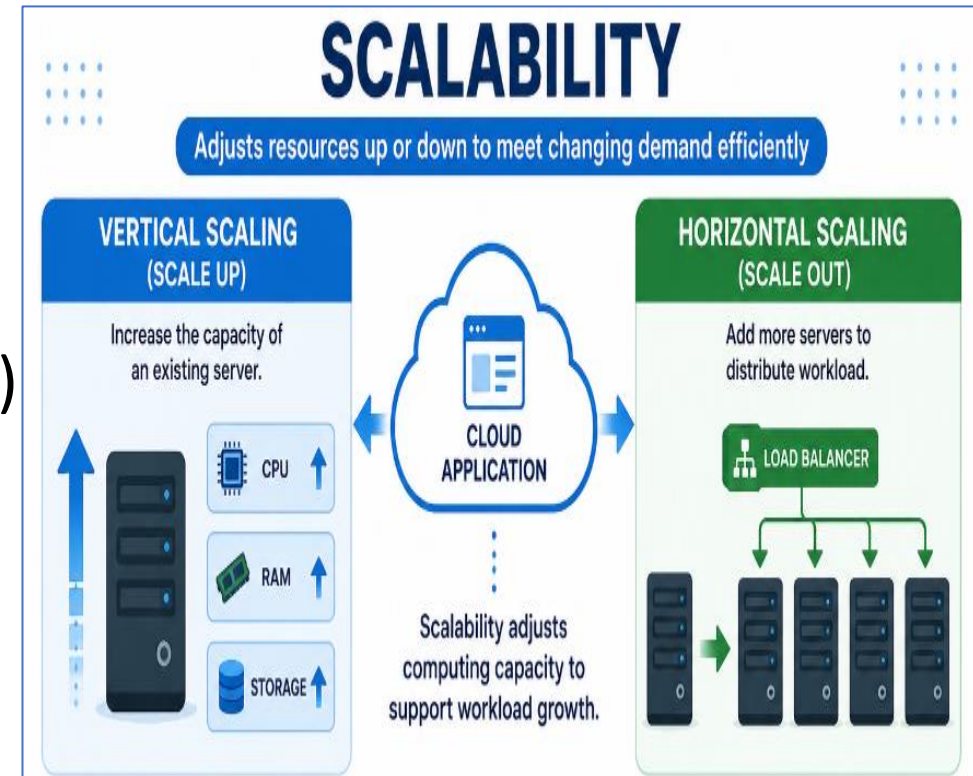
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Cloud Architecture Design Principles

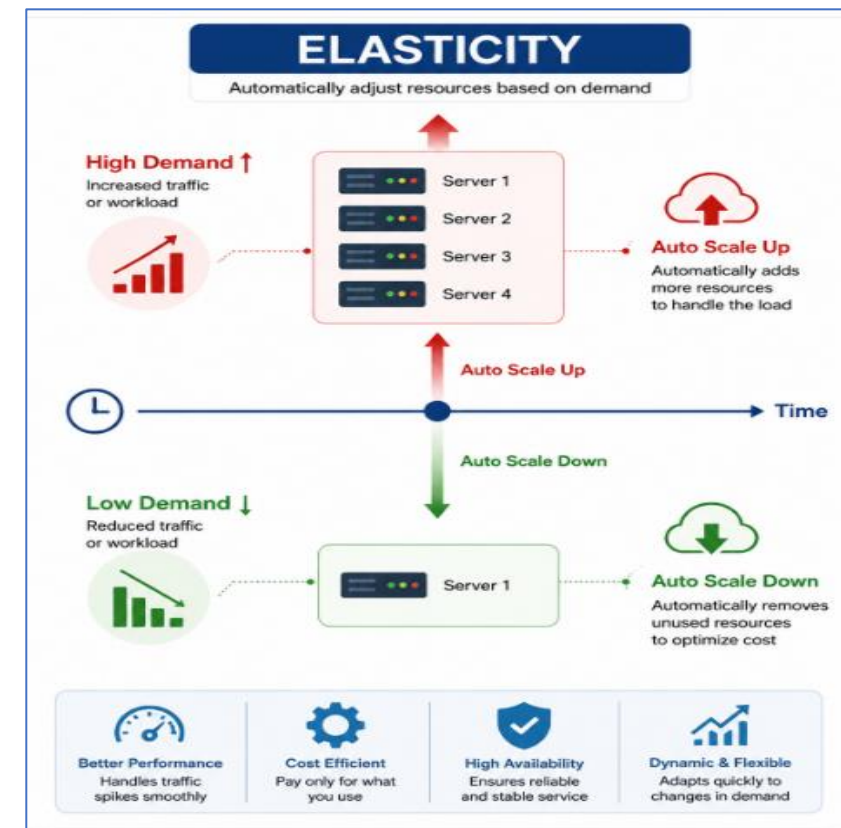
- Each principle contributes to making cloud apps reliable ,efficient and cost-effective.



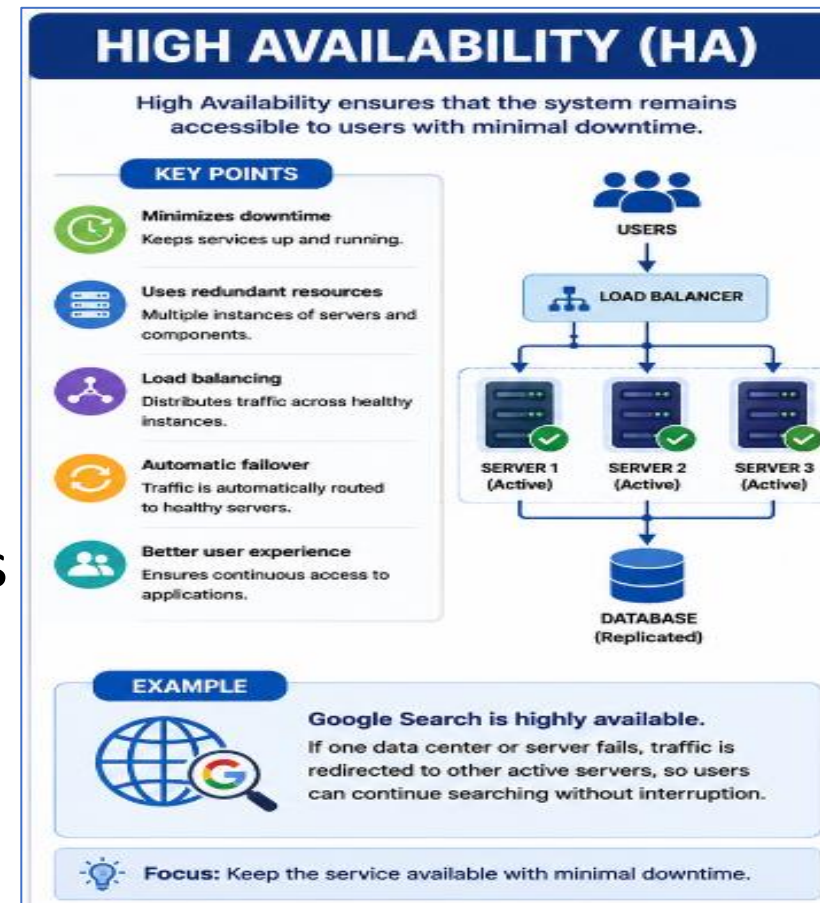
- Scalability is the capability of a cloud system to adjust its resources up or down to meet changing demand efficiently. Increases or decreases computing capacity.
- ✓ Supports workload growth
- ✓ Vertical (**Scale Up**) or Horizontal (**Scale Out**)
- ✓ Maintains application performance
- ✓ Can be manual or automated
- ✓ Preferred in cloud environments for high availability .



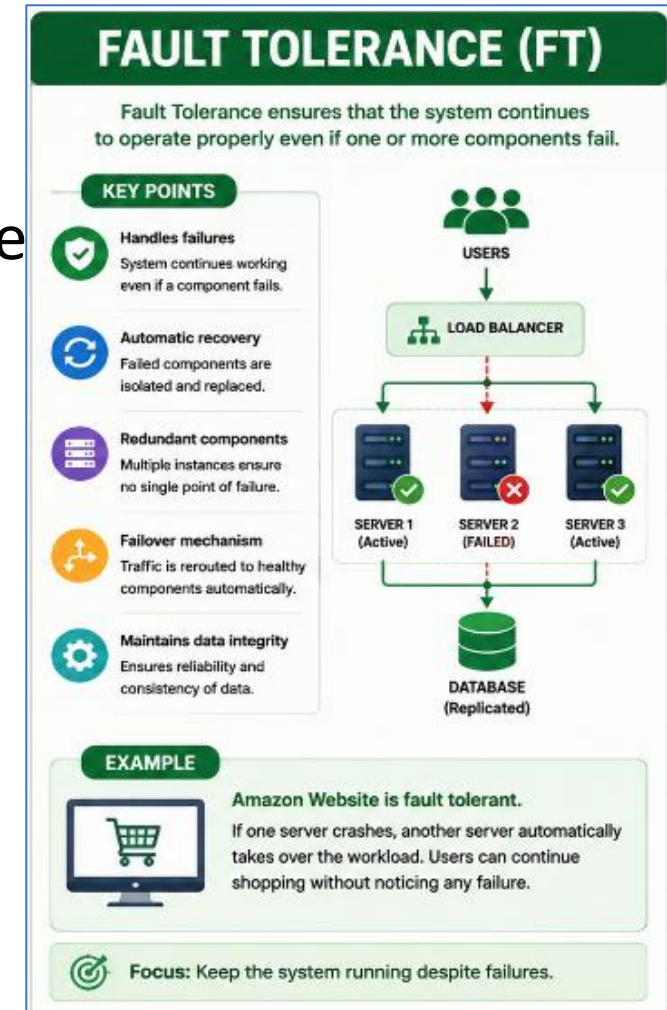
- Elasticity is the capability of a cloud system to automatically allocate or release resources based on real-time workload demand.
- ✓ Automatically allocates and releases resources
- ✓ Matches resources with workload demand
- ✓ Reduces operational costs
- ✓ Improves performance during traffic spikes
- ✓ Scales up and down automatically



- **High Availability** is the capability of a cloud system to keep applications and services continuously accessible with minimal downtime.
- ✓ Minimizes planned and unplanned downtime
- ✓ Keeps applications continuously available
- ✓ Uses redundant servers and load balancing
- ✓ Automatically redirects traffic to healthy servers
- ✓ Improves reliability and user experience
- ✓ Ideal for business-critical cloud applications



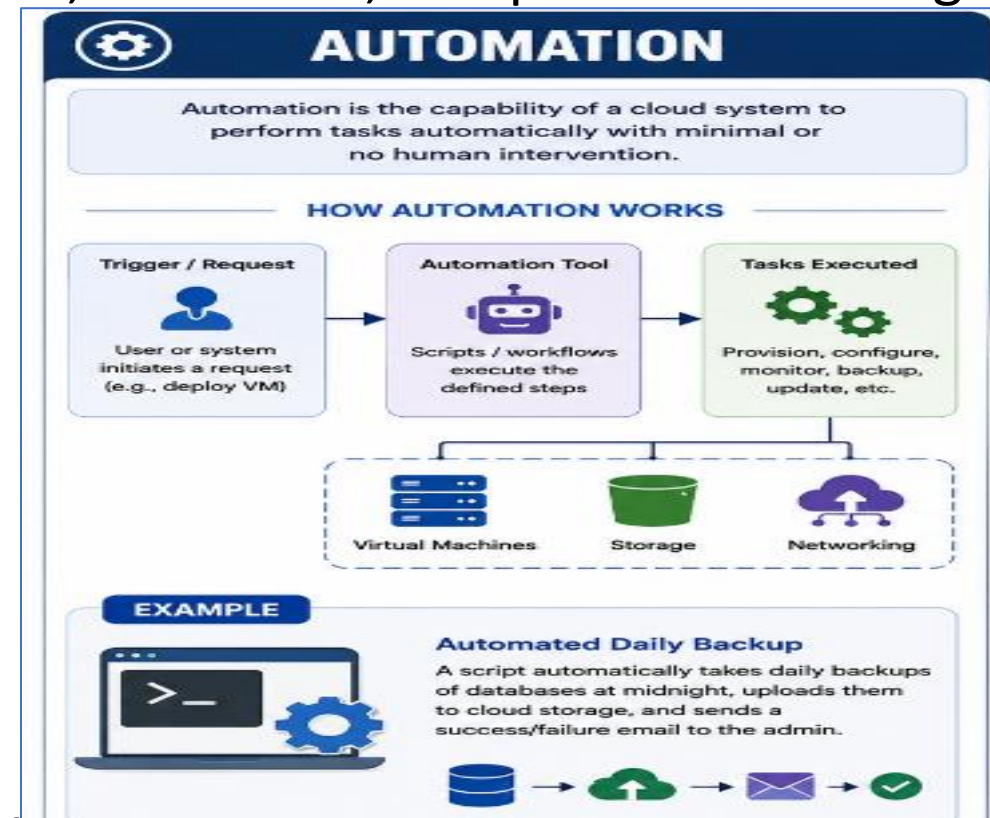
- **Fault Tolerance** is the capability of a cloud system to continue operating even when one or more components fail.
- ✓ Continues operation despite hardware/software failure
- ✓ Detects and isolates failed components
- ✓ Automatically switches to backup resources (failover)
- ✓ Maintains data integrity and service continuity
- ✓ Prevents single points of failure
- ✓ Essential for mission-critical applications



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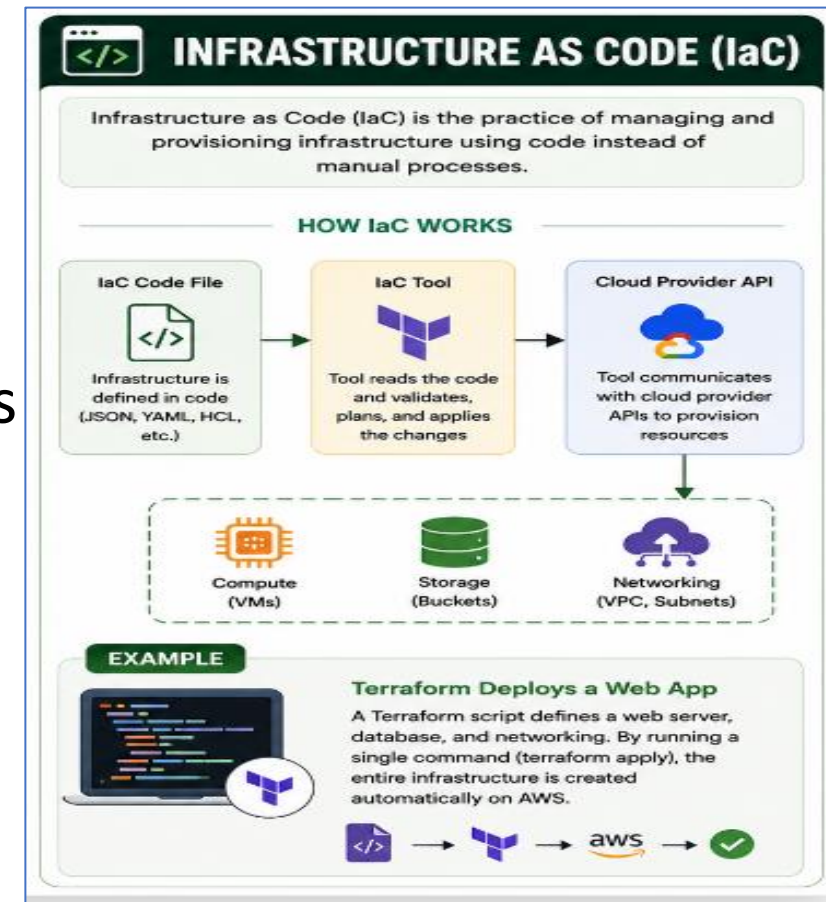
Design Principles: Automation

- **Automation** is the capability of a cloud system to perform tasks automatically with minimal or no human intervention. Uses scripts, workflows, and policies to manage cloud resources efficiently.
- ✓ Eliminates repetitive manual tasks
- ✓ Speeds up resource provisioning
- ✓ Reduces human errors
- ✓ Improves operational efficiency
- ✓ Ensures consistent configurations and Enables faster application deployment

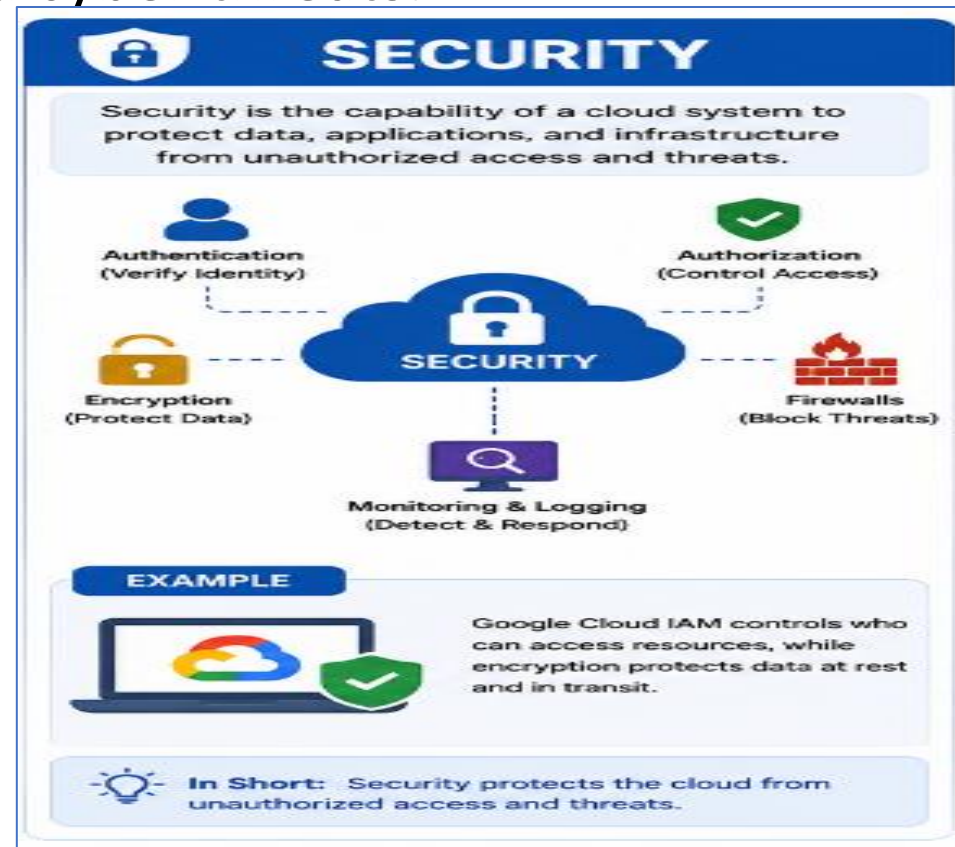


Design Principles: Infrastructure as Code (IaC)

- **Infrastructure as Code (IaC)** is the practice of managing and provisioning cloud infrastructure using code instead of manual configuration. Infrastructure is defined in configuration files and deployed automatically.
- ✓ Automates infrastructure deployment
- ✓ Ensures consistent and repeatable configurations
- ✓ Enables version control and collaboration
- ✓ Simplifies infrastructure management
- ✓ Supports rapid and reliable cloud deployments



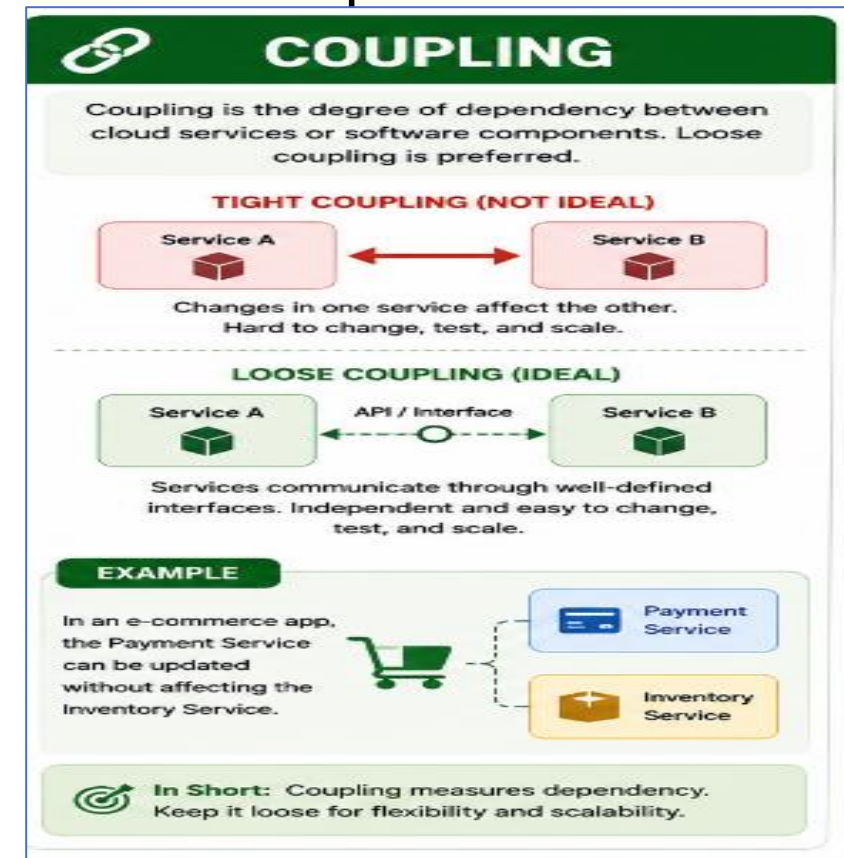
- **Security** is the capability of a cloud system to protect data, applications, and infrastructure from unauthorized access and cyber threats.
- ✓ Protects data confidentiality and privacy
- ✓ Prevents unauthorized access
- ✓ Uses encryption for data protection
- ✓ Supports authentication and authorization
- ✓ Monitors and detects security threats
- ✓ Ensures compliance with security standards



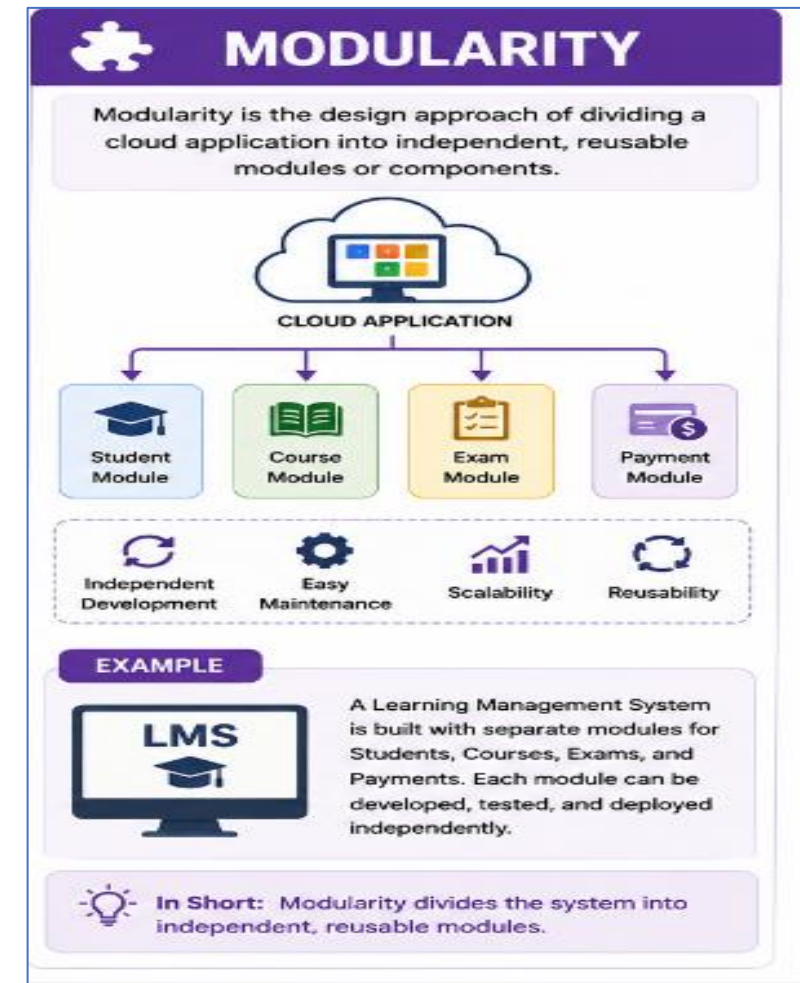
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Design Principles: Coupling

- **Coupling** is the degree of dependency between cloud services or software components. Loose coupling is preferred because services operate independently.
- ✓ Reduces dependency between services
- ✓ Improves flexibility and maintainability
- ✓ Allows independent deployment and updates
- ✓ Increases fault isolation
- ✓ Supports microservices architecture



- **Modularity** is the design approach of dividing a cloud application into independent, reusable modules or components.
- ✓ Simplifies application development
- ✓ Promotes code reuse
- ✓ Makes maintenance easier
- ✓ Supports independent testing and deployment
- ✓ Improves scalability and flexibility
- ✓ Enables faster feature updates



Cloud Concept	Description
 Scalability	Increases system capacity to handle more workload.
 Elasticity	Automatically adjusts resources as demand changes.
 High Availability	Keeps the system running with minimal downtime.
 Fault Tolerance	Keeps working even when components fail.
 Automation	Performs tasks automatically with minimal human effort.
 IaC	Manages infrastructure using code and templates.
 Security	Protects data, applications, and infrastructure.
 Coupling	Measures dependency between system components.
 Modularity	Divides applications into reusable components.

Session Summary

- ✓ Scalability and Elasticity
- ✓ High Availability and Fault Tolerance
- ✓ Automation and Infrastructure as Code (IaC)
- ✓ Security by Design , Loose Coupling, Modularity

Review Questions

1. What is cloud architecture?
2. Explain the importance of architecture design principles.
3. Differentiate between scalability and elasticity.
4. What is High Availability? Define Fault Tolerance with an example.
5. What is Infrastructure as Code (IaC)?
6. Explain loose coupling.
7. List the five pillars of a Well-Architected Framework.

Q1	Q2	Q3	Q4	Q5
Which principle enables applications to handle increasing workloads? A. Automation B. Scalability C. Encryption D. Monitoring	Which principle allows cloud systems to continue operating after hardware failure? A. Elasticity B. Modularity C. Fault Tolerance D. Scalability	Infrastructure as Code (IaC) is used to: A. Write application programs only B. Automate infrastructure deployment C. Design databases manually D. Monitor network traffic	Which cloud principle focuses on reducing operational expenses? A. Reliability B. Cost Optimization C. Availability D. Fault Isolation	Which architecture principle divides applications into independent components? A. High Availability B. Elasticity C. Modularity D. Encryption



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Cloud Computing Life Cycle

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At the end of this session, students will be able to:

- Define the Cloud Computing Life Cycle.
- Explain each phase of the cloud life cycle.
- Understand cloud planning, migration, deployment, operation, and optimization.
- Analyze cloud adoption in real-world organizations.
- Apply cloud life cycle concepts to business scenarios.

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Cloud Computing Life Cycle

Key Points : A typical cloud life cycle consists of the following phases:

- Strategy and Planning
- Assessment
- Migration
- Deployment
- Operations and Management
- Monitoring and Optimization

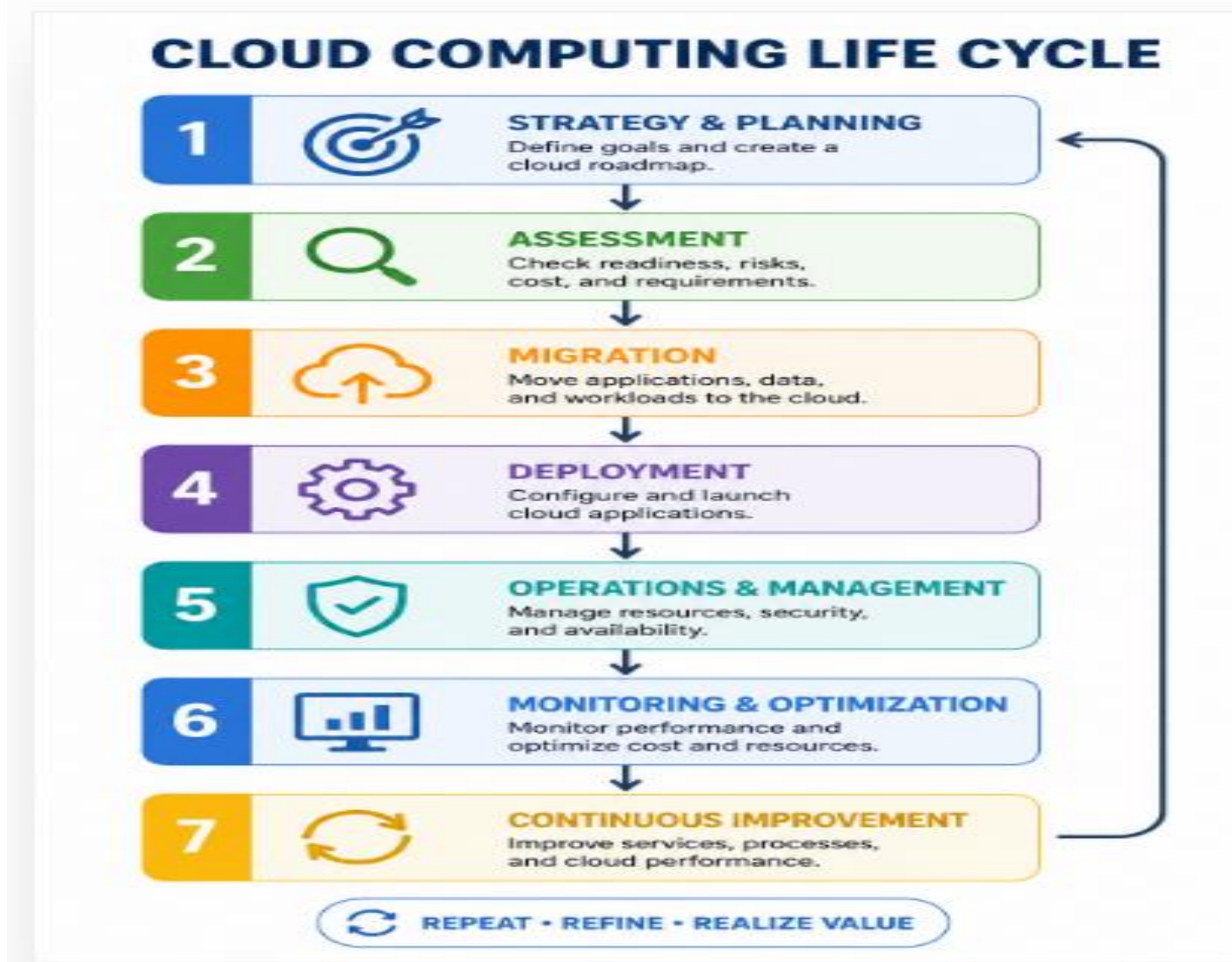
- The **Cloud Computing Life Cycle** is a structured approach that organizations follow when adopting, deploying, managing, and improving cloud-based solutions. It ensures that cloud migration is systematic, cost-effective, secure, and aligned with business goals.
- Unlike simply moving applications to the cloud, the life cycle involves careful planning, assessment, migration, deployment, monitoring, and continuous optimization.

Importance

- Reduces migration risks.
- Minimizes downtime.
- Optimizes costs.
- Enhances security.
- Improves business continuity.
- Ensures efficient resource utilization.

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Cloud Computing Life Cycle



- **Phase1** : The first phase focuses on understanding **why** an organization wants to adopt cloud computing and defining clear business objectives.

- **Activities: Goals → Requirements → Budget → Timeline**

Identify applications, workloads, and business priorities.

Select suitable cloud and service models.

Benefit: Reduces risks and improves resource planning.

Example: A university plans to migrate its LMS to the cloud.

- **Phase2** : The assessment phase evaluates whether existing applications, databases, and infrastructure are suitable for cloud migration.
- **Activities: Compatibility → Security → Cost → Compliance**
Identify technical dependencies and migration challenges.
Classify workloads as migrate, modify, or retain.
Benefit: Reduces migration failures and unexpected costs.
Example: A hospital assesses patient-data applications before migration.

- **Phase3** : Cloud migration is the process of transferring applications, data, and workloads from on-premises infrastructure to the cloud.
- **Activities: Plan → Transfer → Validate → Complete**
Choose suitable migration methods for each workload.
Test data integrity and application functionality.
Benefit: Enables a smooth transition with minimal disruption.
Example: A company moves its database from an on-premises server to Cloud SQL.

- **Phase4** : Manage cloud resources and applications after deployment.
- **Configure → Test → Release → Scale**

Set up networking, security, storage, and compute resources.

Validate application performance before full release.

Benefit: Ensures reliable and secure application availability.

Example: A web application is deployed on cloud VMs behind a load balancer.

- **Phase5** : Once applications are deployed, organizations must ensure they continue to operate efficiently.
- **Activities:Manage → Secure → Backup → Support**
Monitor resources, users, configurations, and service availability.
Apply updates, backups, and security controls regularly.
Benefit: Maintains reliable and secure cloud operations.
Example: IT staff manage VM instances, IAM permissions, and backups.

- **Phase6** : Monitor performance and improve resource efficiency.

- **Activities: Monitor → Analyse → Adjust → Optimise**

Track CPU, memory, network usage, and application performance.

Resize or scale resources based on workload requirements.

Benefit: Improves performance while controlling cloud costs.

Example: Automatically adding VMs when website traffic increases.

- **Phase7** : Continuously enhance cloud services based on usage and feedback.
- **Activities: Review → Improve → Automate → Repeat**
Identify performance, security, cost, and usability improvements.
Adopt new cloud technologies and better practices.
Benefit: Keeps the cloud environment efficient and future-ready.
Example: A company continuously optimizes its cloud architecture to reduce costs and improve application performance.

Session Summary

- ✓ Cloud Computing Life Cycle
- ✓ Strategy and Planning
- ✓ Assessment
- ✓ Migration
- ✓ Deployment
- ✓ Operations and Management
- ✓ Monitoring and Optimization
- ✓ Continuous Improvement

Review Questions

1. Define the Cloud Computing Life Cycle.
2. Explain the importance of planning before cloud migration.
3. What activities are performed during the assessment phase?
4. Differentiate between Rehost and Refactor migration strategies.

1. Which is the first phase of the cloud life cycle? A. Migration B. Strategy and Planning C. Deployment D. Monitoring	2. Which migration strategy is also known as "Lift and Shift"? A. Refactor B. Rehost C. Repurchase D. Retire	3. Which phase focuses on improving cloud performance and reducing costs? A. Assessment B. Deployment C. Monitoring and Optimization D. Planning
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Cloud Reference Architecture

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At the end of this session, students will be able to:

- Define Cloud Reference Architecture (CRA).
- Explain the need for a reference architecture.
- Identify the major components of a cloud reference architecture.
- Understand the roles of cloud consumers, providers, brokers, auditors, and carriers.
- Explain data flow and service interaction in a cloud environment.
- Apply reference architecture concepts to real-world cloud platforms.

- **Key Points : Why is Cloud Reference Architecture Important?**
- Provides a standardized design.
- Simplifies cloud implementation.
- Ensures interoperability between services.
- Improves scalability and security.
- Reduces design complexity.
- Supports best practices.

Example

When a company builds an online shopping platform on Google Cloud or AWS, it follows architectural guidelines for networking, storage, security, databases, and application deployment rather than designing everything independently.

- A **Cloud Reference Architecture (CRA)** is a standardized blueprint that describes the structure, components, relationships, and interactions within a cloud computing environment. It provides a common framework for designing, deploying, and managing cloud-based systems.
- Instead of creating cloud solutions from scratch, organizations use a reference architecture as a guide to ensure that systems are scalable, secure, reliable, and interoperable.
- The **National Institute of Standards and Technology (NIST)** has proposed a widely accepted Cloud Reference Architecture that defines the major actors,

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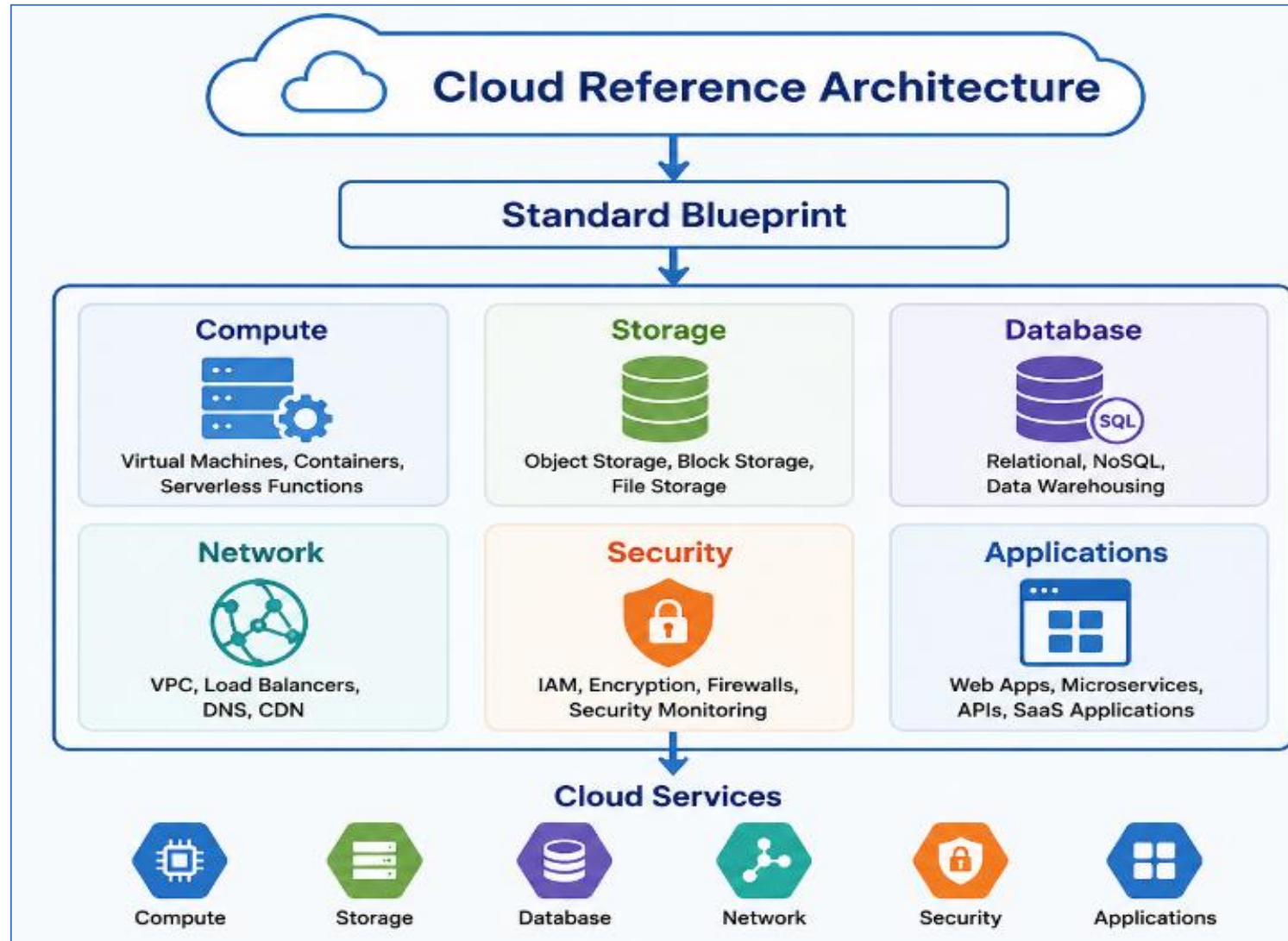
Cloud Reference Architecture Introduction

- **CRA and NIST :**
 - Cloud Reference Architecture = Technology & Components (What?)
 - NIST Reference Architecture = Roles & Responsibilities (Who?)

- The **CRA Cloud Reference Architecture** identifies five primary components involved in cloud computing.
- **Front End** – Provides user access through interfaces and applications.
- **Core Services** – Includes compute, storage, databases, and networking.
- **Virtualization** – Enables efficient sharing and allocation of resources.
- **Security & IAM** – Protects resources and controls user access.
- **Monitoring & Management** – Tracks performance and manages cloud resources.
- **Example:** Google Cloud Console manages VMs, storage, networking, IAM, and monitoring from one interface.

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Cloud Reference Architecture Introduction

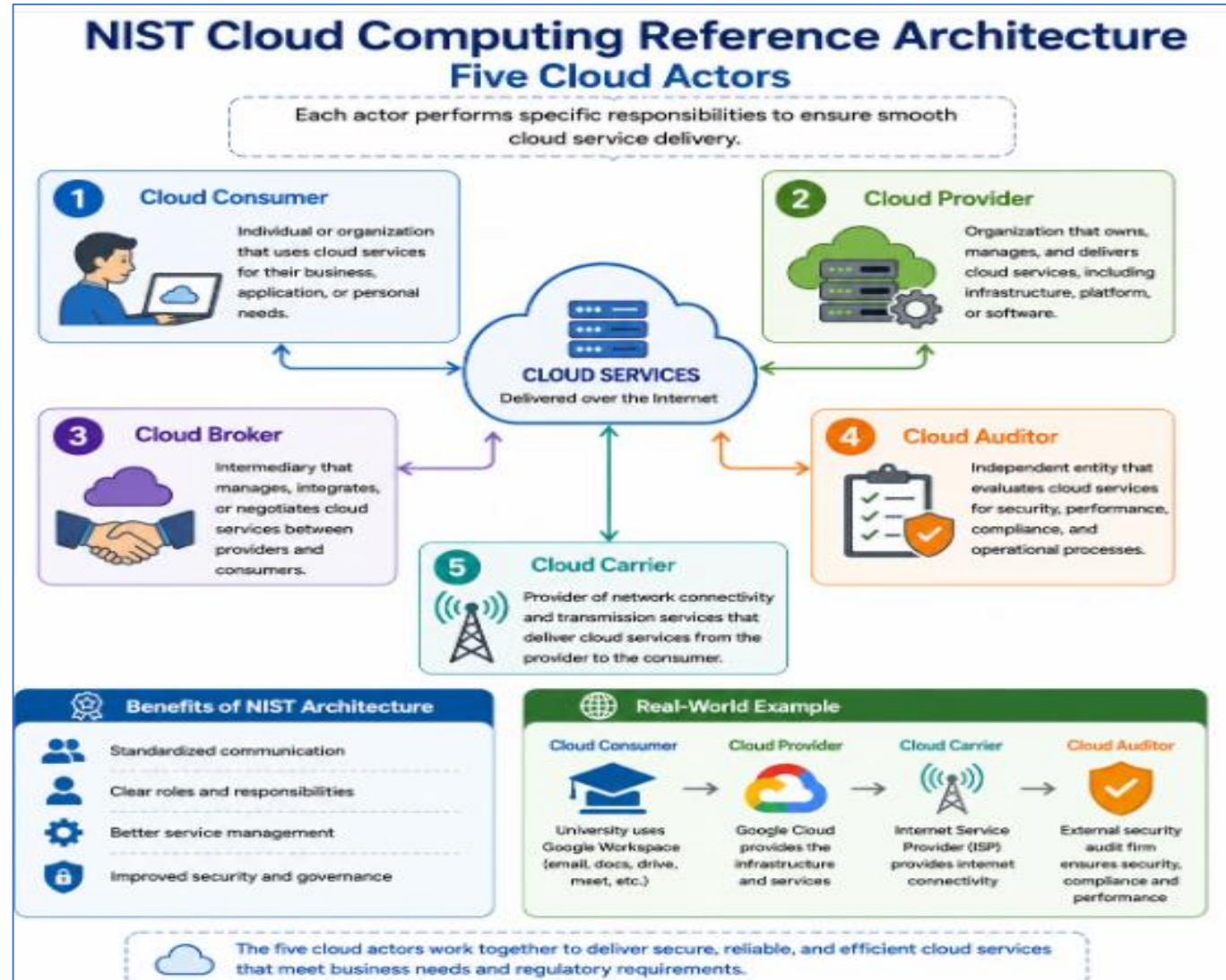


Cloud Reference Architecture (NIST):Actors

- The **NIST Cloud Reference Architecture** identifies five primary actors involved in cloud computing.
- **Cloud Consumer** – Uses cloud services and resources.
- **Cloud Provider** – Delivers and manages cloud services.
- **Cloud Broker** – Integrates or manages services between providers and consumers.
- **Cloud Auditor** – Checks security, performance, and compliance.
- **Cloud Carrier** – Provides network connectivity to access cloud services.
- **Example:** University uses Google Workspace; Google provides services, ISP connects users, and an auditor checks compliance.

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Cloud Reference Architecture Components:NIST



- A Cloud services interact continuously to process user requests.
1. **Request** – User sends a request to the cloud.
 2. **Authentication** – User identity is verified.
 3. **Load Balancing** – Request is routed to an available server.
 4. **Processing** – Application server processes the request.
 5. **Data Access** – Database/storage provides the required data.
 6. **Response** – Processed data is securely returned to the user.
- **Example : When opening Google Drive:**
 - User login is authenticated.
 - Storage service retrieves the requested files.
 - Network delivers the data.
 - User views and accesses the documents.

Cloud Reference Architecture – Google Cloud Example

- Google Cloud follows reference architecture principles through its integrated services.
 1. **Access** – Cloud Console provides centralized user management.
 2. **Compute** – Compute Engine runs applications and workloads.
 3. **Storage & Database** – Cloud Storage stores files; Cloud SQL manages structured data.
 4. **Networking** – VPC and Load Balancing provide secure and reliable connectivity.
 5. **Security & Monitoring** – IAM controls access and Cloud Monitoring tracks performance.
- **Workflow:** User → Internet → Load Balancer → Compute Engine → Cloud SQL/Storage → Response.

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Session Summary & Review Questions

Session Summary

- ✓ Cloud Reference Architecture
- ✓ NIST Cloud Reference Architecture
- ✓ Cloud Consumer , Cloud Provider, Cloud Broker, Cloud Carrier, Cloud Auditor
- ✓ Cloud Components
- ✓ Cloud Service Interaction
- ✓ Google Cloud Reference Architecture

Review Questions

1. Define Cloud Reference Architecture.
2. Why is a reference architecture important?
3. Explain the five actors in the NIST Cloud Reference Architecture.
4. Differentiate between a Cloud Provider and a Cloud Broker.
5. What is the role of a Cloud Auditor?
6. Explain the workflow of cloud service interaction.
7. Describe the major components of a cloud reference architecture.

1. Which organization proposed the widely accepted Cloud Reference Architecture?

- A. IEEE
- B. NIST**
- C. ISO
- D. W3C

2. Which actor provides Internet connectivity for cloud services?

- A. Cloud Broker
- B. Cloud Carrier**
- C. Cloud Auditor
- D. Cloud Consumer

3. Which component verifies user identity before accessing cloud services?

- A. Storage
- B. Identity and Access Management (IAM)**
- C. Compute Engine
- D. Cloud Broker



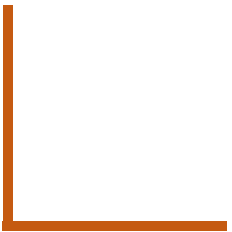
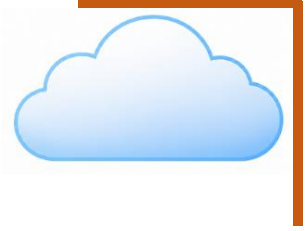
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Cloud Reference Architecture

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At the end of this session, students will be able to:

- Define load balancing and explain its importance in cloud computing.
- Describe the working of a load balancer.
- Differentiate between static and dynamic load balancing.
- Explain common load balancing algorithms.
- Understand hardware and software load balancers.
- Analyze real-world cloud load balancing scenarios.

- **What is Load Balancing?**
- **Load balancing** is the process of distributing incoming network traffic or user requests across multiple servers to ensure that no single server becomes overloaded. It improves application performance, reliability, and availability.
- In cloud computing, thousands or even millions of users may access an application simultaneously. A load balancer acts as a **traffic manager**, intelligently directing requests to available servers.

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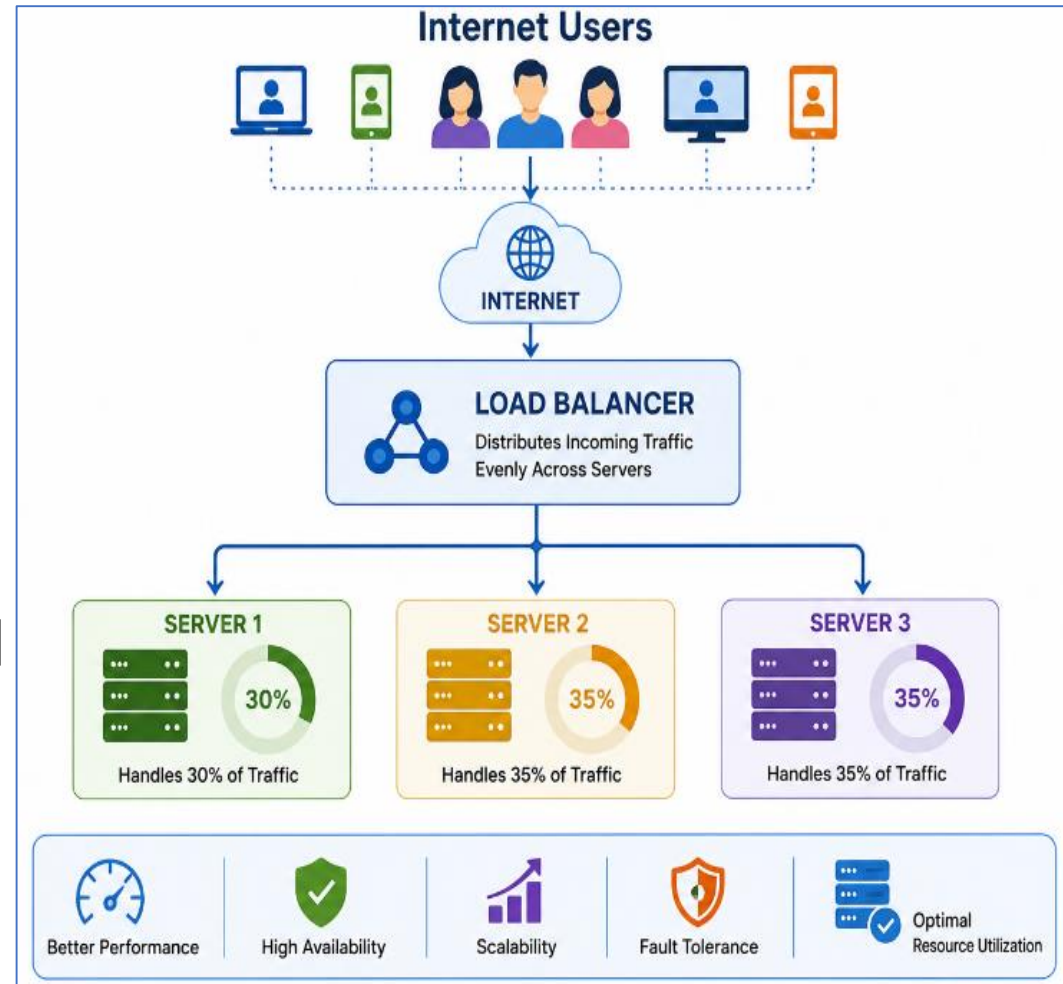
Cloud Load Balancing:

- **Importance of Load Balancing**
- **Purpose** – Distributes workloads across multiple servers.
- **Prevents Overload** – Avoids excessive traffic on a single server.
- **Performance** – Improves response time and resource utilization.
- **Availability** – Keeps applications available during server failures.
- **Scalability** – Supports increasing numbers of users and requests.
- **Example:** An e-commerce load balancer distributes festive-sale traffic across multiple web servers

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Cloud Load Balancer: Working

- A load balancer sits between clients and application servers. It receives incoming requests and forwards each request to the most appropriate server based on predefined rules or real-time server conditions.



Category	Details
Steps in Load Balancing	1. User sends a request. 2. The request reaches the load balancer 3. The load balancer checks server availability and health. 4. It selects the best or least-busy server. 5. The request is forwarded to the selected server. 6. The server processes the request. 7. The response is sent back to the user.
Benefits	• Faster response time. • Efficient resource utilization. • Reduced server downtime. • Improved user experience.
Example	When users access Google Search , incoming requests are automatically routed by load balancers to the nearest and least-busy servers, ensuring fast search results and high availability.

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Cloud Load Balancing: Types

- Load balancing approaches are broadly classified into **Static** and **Dynamic**.

Feature	Static Load Balancing	Dynamic Load Balancing
Definition	Decisions are made before execution using predefined rules.	Decisions are made during execution based on the current system status.
Decision Method	Uses predefined rules.	Continuously monitors server status and workload.
Considers Current Load	No	Yes
Monitoring	Not required	Continuously monitors CPU, memory, response time, etc.
Suitable For	Predictable and stable workloads.	Dynamic cloud environments with changing workloads.
Advantages	- Simple implementation. - Low overhead.	- Better resource utilization. - Improved performance. - Higher reliability.
Disadvantages	Cannot adapt to changing workloads.	- More complex to implement. - Requires continuous monitoring.

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Cloud Load Balancing: Common Algorithms

- A **load balancing algorithm** determines how requests are distributed among servers.

Algorithm	Description	Best Suited For
Round Robin	• Requests are assigned sequentially to each server. • Simple and widely used. • Best when all servers have similar capacity.	Applications with equal-capacity servers and predictable workloads.
Least Connections	• Sends requests to the server with the fewest active connections. • Balances workloads dynamically based on current server usage.	Applications with varying workloads and long-lived user sessions.

Algorithm	Description	Best Suited For
Weighted Round Robin	• Assigns more requests to powerful servers. • Weights are based on server capacity (CPU, memory, etc.).	Environments with servers of different capacities.
IP Hash	• Uses the client's IP address to select a server. • Ensures the same user is routed to the same server when required (session persistence).	Applications requiring user session consistency, such as online banking or shopping carts.
Applications	Common uses of load balancing: • Web hosting • Streaming platforms • E-commerce websites • Online banking systems	Ensures high availability, scalability, and reliable service delivery.

Cloud Load Balancing: Hardware vs Software Load Balancers

- **Hardware Vs Software** Load balancer differences.

Feature	Hardware Load Balancer	Software Load Balancer
Definition	A dedicated physical device used for distributing network traffic across multiple servers.	A software application installed on servers or cloud platforms to distribute traffic.
Examples	• F5 BIG-IP • Citrix ADC (NetScaler) • A10 Networks	• NGINX • HAProxy • Google Cloud Load Balancing • AWS Elastic Load Balancer (ELB)

Cloud Load Balancing: Hardware vs Software Load Balancers

Feature	Hardware Load Balancer	Software Load Balancer
Advantages	• High performance • Highly reliable • Suitable for large enterprises	• Cost-effective • Flexible configuration • Easily scalable
Disadvantages	• Expensive • Limited scalability compared to cloud-based solutions	• Depends on server resources • Performance depends on underlying infrastructure
Best Suited For	Large enterprise data centers with high-volume traffic.	Cloud environments, web applications, and small to medium-sized businesses requiring scalable load balancing.

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Cloud Load Balancing: Hardware vs Software Load Balancers

TYPES OF LOAD BALANCER

HARDWARE LOAD BALANCER

A dedicated physical device used for distributing traffic.



ADVANTAGES

- High performance
- Reliable
- Suitable for large enterprises



DISADVANTAGES

- Expensive
- Limited scalability

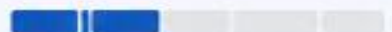


BEST SUITED FOR

Large enterprise data centers with high-volume traffic.



SCALABILITY

 Limited



COST

 High

VS

SOFTWARE LOAD BALANCER

A software application installed on servers or cloud platforms.



ADVANTAGES

- Cost-effective
- Flexible
- Easily scalable



DISADVANTAGES

- Depends on server resources



BEST SUITED FOR

Cloud environments, web applications, and small to medium businesses.



SCALABILITY

 High



COST

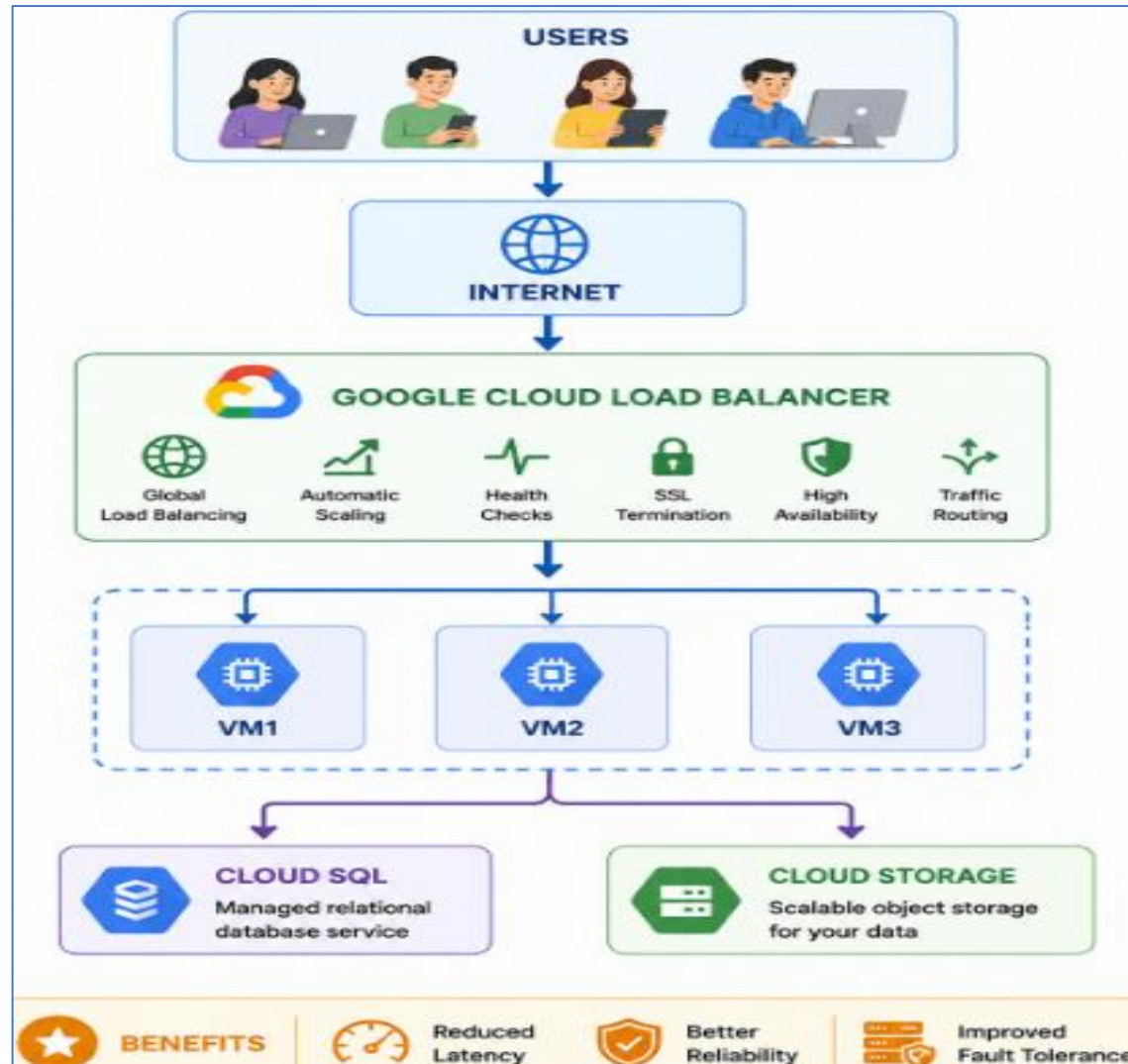
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Google Cloud provides **Cloud Load Balancing**, a fully managed service that distributes user traffic across multiple virtual machines and regions.

Category	Details
Features	• Global load balancing • Automatic scaling • Health checks • SSL termination • High availability • Intelligent traffic routing
Workflow	1. User sends a request.2. Cloud Load Balancer receives the incoming traffic.3. Health checks identify healthy and available VM instances.4. The request is forwarded to the healthiest or nearest VM instance.5. The processed response is returned to the user.
Benefits	• Reduced latency • Better reliability • Improved fault tolerance
Real-World Example	A global e-commerce company hosts its website on Google Cloud. During festive sales, Cloud Load Balancing automatically distributes millions of customer requests across multiple regions, ensuring high availability, low latency, and uninterrupted service.

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Cloud Load Balancing: Google Cloud Platform



Session Summary

- ✓ Introduction to Load Balancing
- ✓ Working of a Load Balancer
- ✓ Static Load Balancing and Dynamic Load Balancing
- ✓ Load Balancing Algorithms
- ✓ Hardware and Software Load Balancers
- ✓ Google Cloud Load Balancing

Review Questions

1. Define load balancing.
2. Why is load balancing important in cloud computing?
3. Explain the working of a load balancer.
4. Differentiate between static and dynamic load balancing.
5. Describe the Round Robin algorithm.
6. What is the Least Connections algorithm?
7. Compare hardware and software load balancers.
8. Explain Google Cloud Load Balancing.
9. How does load balancing improve fault tolerance?

1. Which component distributes incoming requests among multiple servers? A. Firewall B. Load Balancer C. Database D. Hypervisor	2. Which load balancing algorithm sends requests sequentially to each server? A. Least Connections B. Round Robin C. IP Hash D. Random	3. Which type of load balancing continuously monitors server status? A. Static B. Dynamic C. Manual D. Sequential
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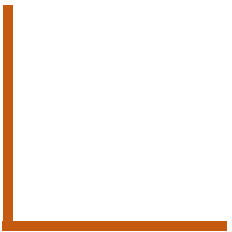
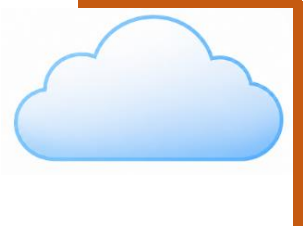
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Cloud Virtualization Technology

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At the end of this session, students will be able to:

- Define virtualization and explain its importance in cloud computing.
- Describe the architecture and working of virtualization.
- Explain the role of hypervisors.
- Differentiate between physical and virtual environments.
- Identify the advantages and challenges of virtualization.
- Analyze real-world applications of virtualization in cloud platforms.

CLOUD TECHNOLOGIES

Cloud Virtualization Technology Introduction

- **Virtualization** is the technology that enables multiple virtual computers, called **Virtual Machines (VMs)**, to run on a single physical computer. Each virtual machine behaves like an independent computer with its own operating system, applications, CPU, memory, storage, and network interface.

CLOUD TECHNOLOGIES

Cloud Virtualization Technology Introduction

- Virtualization is the **foundation of cloud computing** because it allows cloud providers to efficiently utilize physical hardware while delivering isolated computing environments to multiple users.
- Instead of assigning one physical server to one application, virtualization enables a single server to host multiple virtual machines, improving resource utilization and reducing costs.

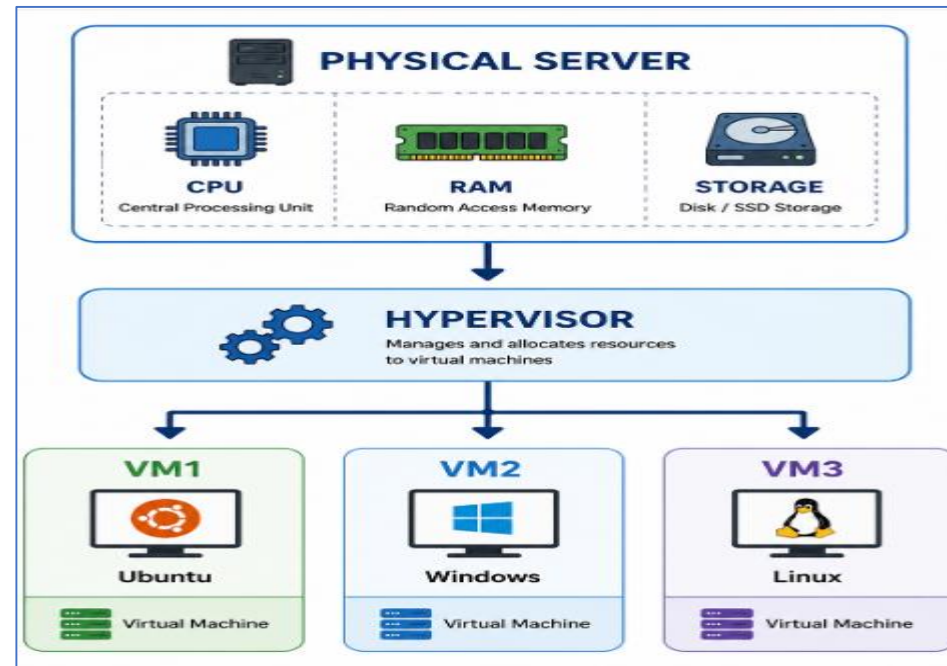
CLOUD TECHNOLOGIES

Cloud Virtualization Technology Introduction

- **Why is Virtualization Important?**
- Efficient use of hardware resources.
- Reduced infrastructure costs.
- Faster deployment of servers.
- Isolation between applications.
- Better scalability.
- Easier backup and disaster recovery.

Real-World Example

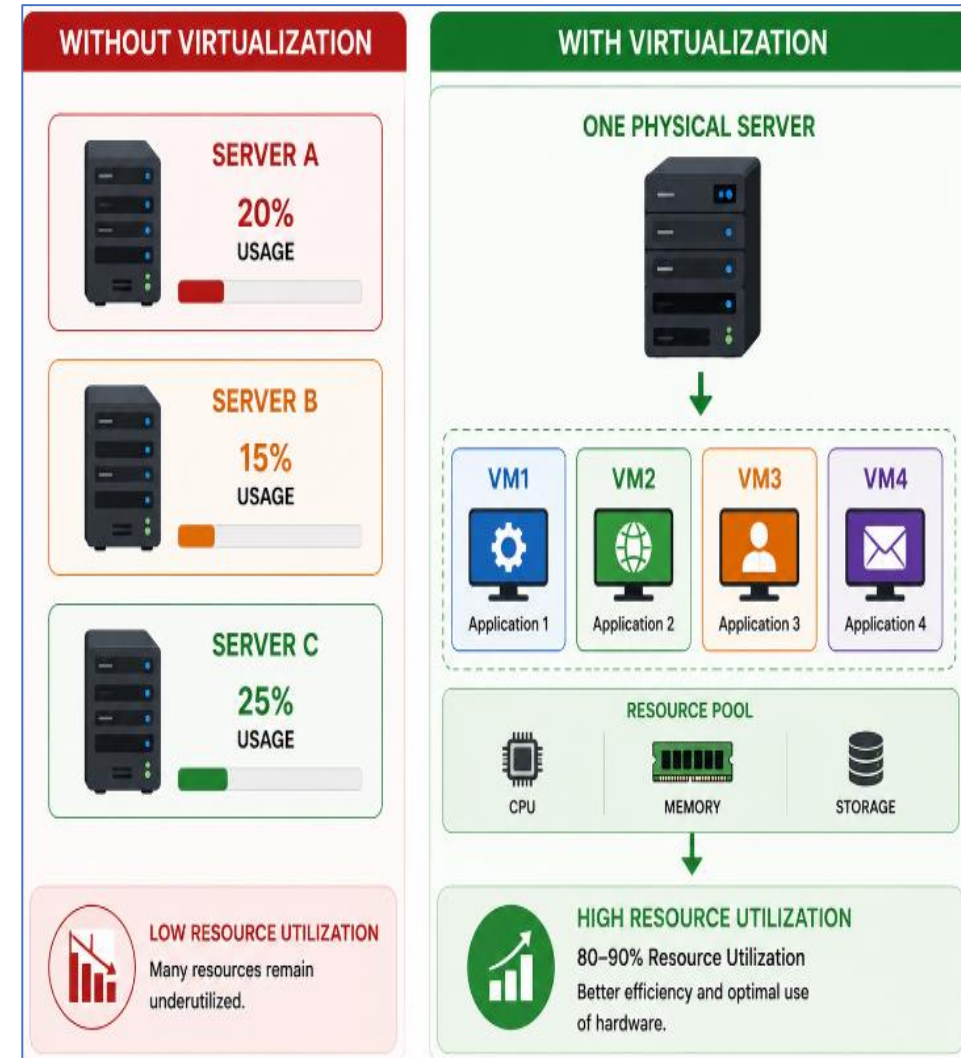
Google Cloud Compute Engine allows users to create virtual machines without purchasing physical servers. Multiple customer VMs run securely on the same physical hardware.



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Cloud Virtualization Need and Benefits of Virtualization

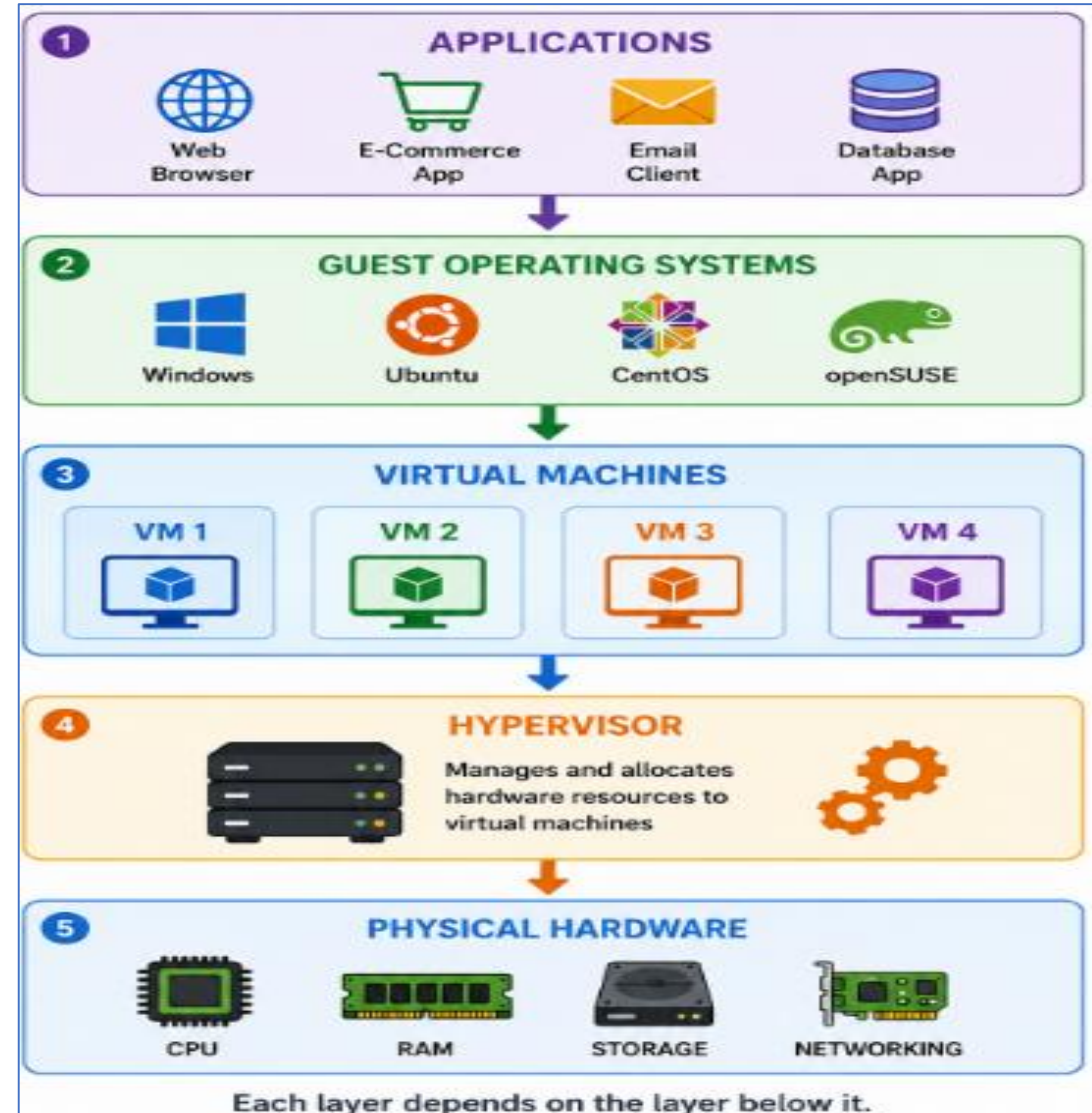
- Before virtualization, organizations often dedicated one physical server to a single application, resulting in low hardware utilization and higher operational costs.
- Virtualization addresses these issues by allowing multiple virtual machines to share the same physical resources.



CLOUD TECHNOLOGIES

Cloud Virtualization Architecture

- **Virtualization architecture**
Consists of several layers working together.
- **Advantages**
 - Better resource sharing.
 - Independent operating systems.
 - Simplified management.
 - Secure isolation between virtual machines



- **Layers**

1. **Physical Hardware** – Provides CPU, RAM, storage, and networking resources.
2. **Hypervisor** – Creates and manages virtual machines by allocating hardware resources.
3. **Virtual Machines (VMs)** – Independent virtual environments that receive allocated resources.
4. **Guest Operating Systems** – Each VM runs its own operating system (e.g., Windows, Linux, Ubuntu).
5. **Applications** – Software and services run inside each guest operating system.

- **Working Process**
 1. **The physical server** provides CPU, RAM, storage, and networking resources.
 2. **The hypervisor** divides and allocates these resources among multiple virtual machines.
 3. **Each virtual machine** receives its assigned resources.
 4. **Every VM runs** its own guest operating system independently.
 5. **Users access** the applications running inside the virtual machines.

Cloud Hypervisors – The Core of Virtualization

- A **Hypervisor**, also called a **Virtual Machine Monitor (VMM)**, is software that creates, manages, and monitors managing multiple virtual machines on a single physical server. Without a hypervisor, virtualization would not be possible.
- **Responsibilities**
 - Create and manage virtual machines (VMs).
 - Allocate CPU, RAM, storage, and networking resources.
 - Isolate virtual machines from one another.
 - Monitor VM performance and resource usage.
 - Manage and optimize resource allocation.
- **Examples:** VMware ESXi, Microsoft Hyper-V, KVM (Kernel-based Virtual Machine), Xen Hypervisor, Oracle VM VirtualBox (Desktop Virtualization)

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Cloud Virtualization vs Physical Computing

- Comparison of Traditional Physical Environment and Virtualized Environment**

Feature	Traditional Physical Environment	Virtualized Environment
Operating Systems	One operating system per physical server.	Multiple operating systems run on a single physical server using virtual machines.
Hardware Usage	Low resource utilization.	High and efficient resource utilization.
Scalability	Limited scalability.	Excellent scalability; VMs can be created, resized, or removed easily.
Infrastructure Cost	Higher hardware, power, cooling, and maintenance costs.	Reduced infrastructure and operational costs.
Resource Sharing	Resources are dedicated to a single operating system.	CPU, RAM, storage, and networking resources are shared among multiple VMs.
Maintenance	More hardware to maintain and manage.	Easier centralized management with fewer physical servers.

- **Virtualization** is widely used across industries.

Category	Details
Major Applications	• Cloud Computing • Software Testing • Disaster Recovery • Data Centres • Educational Laboratories • Virtual Desktop Infrastructure (VDI) • Application Development
Industry Examples	Education: Virtual computer laboratories for students. Healthcare: Hospital information and secure patient data management. Banking: Secure application hosting and high-availability financial services. Software Companies: Development, testing, and QA environments using VM

- Challenges and Best Practices in Virtualization

Category	Details
Challenges	- Resource contention among virtual machines. - Performance overhead due to virtualization. - Security vulnerabilities. - VM sprawl (too many unused or unmanaged virtual machines). - Licensing costs. - Complex management in large environments.
Best Practices	- Monitor CPU, memory, storage, and network usage regularly. - Remove unused virtual machines to avoid VM sprawl. - Apply security patches and updates promptly. - Use backups and VM snapshots for disaster recovery. - Allocate resources appropriately to each VM. - Implement Role-Based Access Control (RBAC) to enhance security.
Example	An organization continuously monitors CPU and memory utilization to prevent performance degradation caused by excessive virtual machines and optimizes resource allocation to maintain efficient system performance.

Session Summary

- ✓ Definition of Virtualization
- ✓ Importance of Virtualization
- ✓ Benefits of Virtualization
- ✓ Virtualization Architecture
- ✓ Hypervisors
- ✓ Physical vs Virtual Environments
- ✓ Applications of Virtualization
- ✓ Challenges and Best Practices

Review Questions

1. Define virtualization.
2. Why is virtualization considered the foundation of cloud computing?
3. Explain the role of a hypervisor.
4. Draw and explain virtualization architecture.
5. Compare physical and virtual computing environments.
6. List five applications of virtualization.
7. Explain the benefits of virtualization.
8. What are the challenges of virtualization?
9. Why is isolation important in virtualization?

1. Which software is responsible for creating and managing virtual machines? A. Operating System B. Hypervisor C. Database D. Firewall	2. Virtualization primarily improves: A. Hardware wastage B. Resource utilization C. Network latency only D. Manual processing	3. Which of the following is an example of a hypervisor? A. MySQL B. VMware ESXi C. Apache HTTP Server D. Google Chrome
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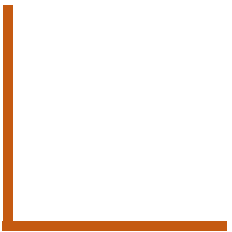
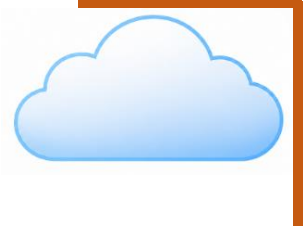
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CLOUD COMPUTING

Cloud Types of Virtualization

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At the end of this session, students will be able to:

- Define different types of virtualization.
- Differentiate between server, storage, network, desktop, application, and OS virtualization.
- Explain Type 1 and Type 2 hypervisors.
- Identify the advantages and applications of each virtualization type.
- Relate virtualization technologies to cloud computing platforms such as Google Cloud.

- **Virtualization** is one of the core technologies that enables cloud computing. It allows a single physical computer or infrastructure to support multiple virtual environments. Depending on the resource being virtualized, virtualization can be classified into several types.
- **Major Types of Virtualization**
 1. Server Virtualization
 2. Storage Virtualization
 3. Network Virtualization

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Cloud Introduction to Types of Virtualization

4. Desktop Virtualization
5. Application Virtualization
6. Operating System (OS-Level) Virtualization

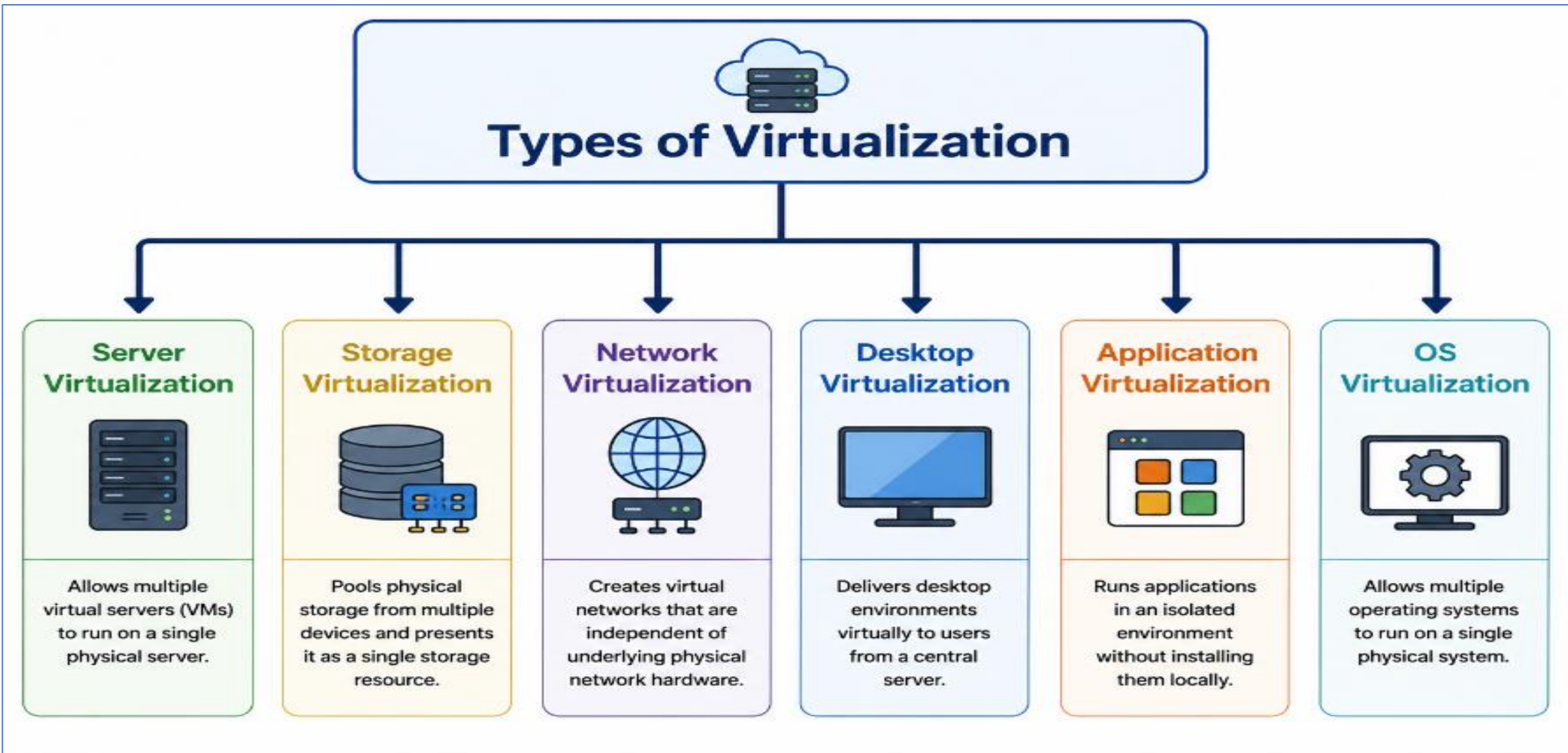
Each type addresses a different business need and helps improve resource utilization, flexibility, and scalability.

Organizations virtualize different resources depending on their requirements.

- A cloud provider virtualizes servers to host virtual machines.
- A bank virtualizes storage to manage customer data efficiently.
- A university uses desktop virtualization to provide virtual computer labs.
- **Example**
- Google Cloud Platform uses **server virtualization** to create virtual machines, **storage virtualization** for Cloud Storage, and **network virtualization** through Virtual Private Cloud (VPC).

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Types of Virtualization



- **Server Virtualization: Runs** multiple virtual machines on one physical server.

Benefit: Improves resource utilization and reduces hardware costs.

Example: GCP Engine creates multiple virtual machines on powerful physical servers.

- **Storage Virtualization: Combines** storage from multiple devices into one logical storage pool.

Benefit: Simplifies storage management and improves utilization.

Example: GCP stores data across multiple disks while presenting a single storage service.

Types of Virtualization

- **Network Virtualization: Creates** virtual networks independent of physical network hardware.

Benefit: Improves flexibility, security, and network management.

Example: Google Cloud VPC(Virtual Private Cloud) enables organizations to create secure virtual networks for applications.

- **Desktop Virtualization: Provides** virtual desktops hosted on a central server or cloud.

Benefit: Enables remote access and centralized management.

Example: A university provides students with virtual desktops containing programming software that can be accessed from anywhere.

Types of Virtualization

- **Application Virtualization** : Runs applications in isolated environments without full local installation.
Benefit: Simplifies deployment and reduces compatibility issues.
Example: GCP runs containerized applications without managing the underlying servers.
- **OS Virtualization – Definition:** Allows multiple isolated operating environments to run on one system.
Benefit: Provides lightweight isolation with efficient resource usage.
Example: Google Kubernetes Engine (GKE) uses containers to deploy and manage cloud applications efficiently.

Cloud Type 1 vs Type 2 Hypervisors

- **Hypervisor** : A hypervisor is software or firmware that creates and manages Virtual Machines (VMs), allowing multiple operating systems to run on the same physical server.

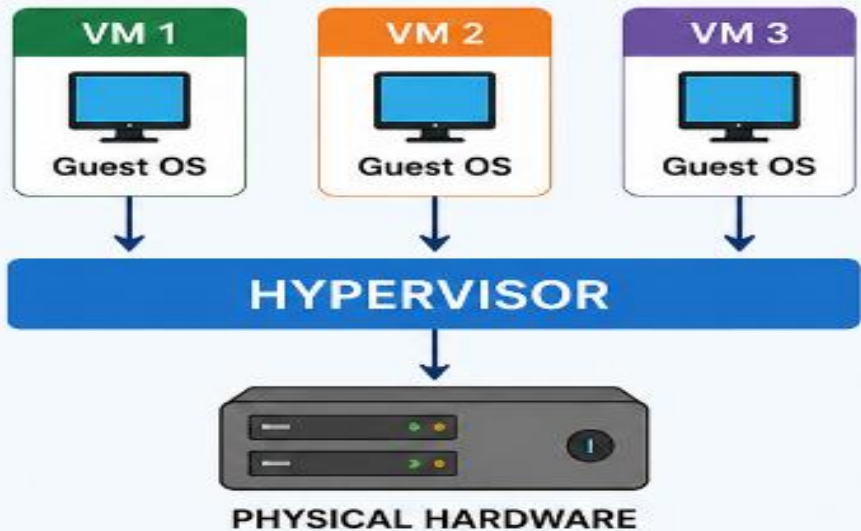
Type	Description	Example
Type 1 – Bare-Metal	Runs directly on physical hardware without a host OS.	VMware ESXi, Microsoft Hyper-V, Xen, .Ideal for servers and data centers
Type 2 – Hosted	Runs on top of a host operating system like a normal application.	Oracle VirtualBox, VMware Workstation, Suitable for development, testing and learning

- **Type 1** = Hardware → Hypervisor → VMs
- **Type 2** = Hardware → Host OS → Hypervisor → VMs
- **Key difference:** Type 1 generally offers better performance and is commonly used in data centers, while Type 2 is convenient for desktop/lab environments.

CLOUD TECHNOLOGIES

Cloud Type 1 vs Type 2 Hypervisors

TYPE 1 HYPERVISOR (BARE-METAL)



Runs directly on physical hardware.



Ideal for servers and data centers.



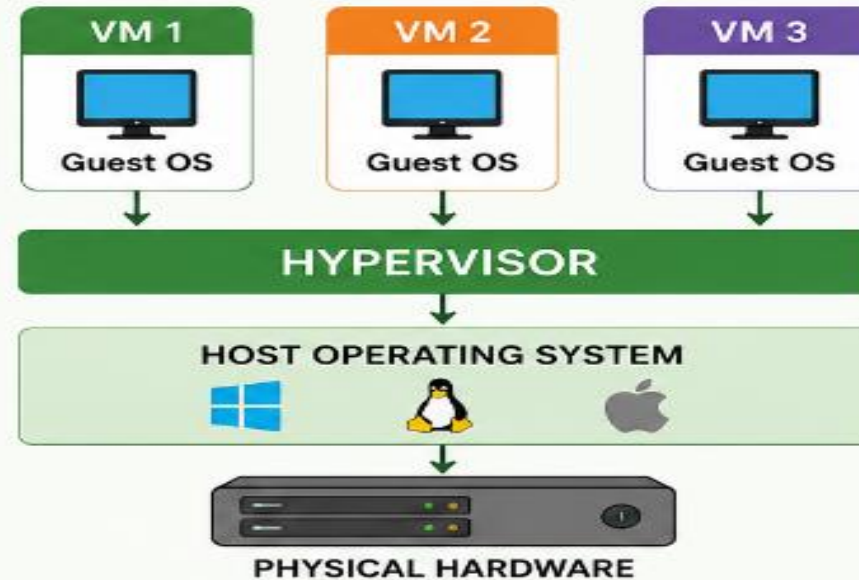
Better performance and security.



Supports many virtual machines.

Examples: VMware ESXi, Microsoft Hyper-V, Xen

TYPE 2 HYPERVISOR (HOSTED)



Runs on top of a host operating system.



Suitable for testing, learning, and development.



Easy to install and use on desktops.



Supports fewer virtual machines.

Examples: Oracle VirtualBox, VMware Workstation

Session Summary

- ✓ Types of Virtualization
- ✓ Server Virtualization
- ✓ Storage Virtualization
- ✓ Network Virtualization
- ✓ Desktop Virtualization
- ✓ Application Virtualization
- ✓ Operating System Virtualization
- ✓ Type 1 and Type 2 Hypervisors

Review Questions

1. Define virtualization.
2. Explain server virtualization with an example.
3. What is storage virtualization?
4. Describe network virtualization.
5. Differentiate between desktop and application virtualization.
6. Explain operating system virtualization.
7. Compare virtual machines and containers.
8. Differentiate between Type 1 and Type 2 hypervisors.

1. Which type of virtualization is used to create multiple virtual servers on one physical server? A. Network Virtualization B. Server Virtualization C. Storage Virtualization D. Desktop Virtualization	2. Which technology is commonly used for operating system virtualization? A. Oracle Database B. Docker C. Apache Server D. MySQL	3. Which hypervisor runs directly on physical hardware? A. Type 2 B. Type 1 C. Hosted Hypervisor D. Desktop Hypervisor
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CLOUD COMPUTING

SOA Components , Communication & Infrastructure

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At the end of this session, students will be able to:

- Define Cloud Service-Oriented Architecture (SOA).
- Explain the need for SOA in cloud computing.
- SOA Components
- SOA Communication
- SOA Infrastructure

Components – What makes up SOA?
Communication – How do the components interact?
Infrastructure – Where and on what platform do they run?

CLOUD TECHNOLOGIES

Cloud SOA: Components

- In cloud computing, **Service-Oriented Architecture (SOA)** consists of independent cloud services that communicate through standard interfaces (APIs).
- Each component performs a specific function and can be developed, deployed, and managed independently.

Major Components

- Cloud Consumer
- Cloud Provider
- API Gateway
- Service Registry
- Microservices
- Cloud Database

Cloud SOA: Components Consumer, Provider

Cloud Consumer

- Requests cloud services.
- Accesses applications through web browsers or mobile apps.
- Uses SaaS, PaaS, or IaaS services.

Cloud Provider

- Delivers cloud services.
- Manages infrastructure.
- Ensures availability and security.

Examples

- Consumer → Student using Google Classroom.
- Provider → Google Cloud, AWS, Microsoft Azure.

An API Gateway is the entry point for all client requests in a cloud application.

Functions

- Routes requests.
- Authentication and authorization.
- Rate limiting.
- Load balancing.
- Request validation and Monitoring.

Cloud Examples

Google API Gateway

AWS API Gateway

Azure API Management

Cloud SOA: Components : Service Registry

A Service Registry stores information about available cloud services and enables service discovery(process of detecting and routing traffic to individual instances of a service within a network).

Functions

- Service registration.
- Service discovery.
- Service lookup.
- Dynamic routing.

Cloud Examples

Kubernetes Service Discovery

Consul

Eureka

Microservices divide an application into small independent services. **Functions**

Characteristics

- Independent deployment.
- Independent scaling.
- API-based communication.
- Fault isolation.

Cloud Examples

Login Service

Payment Service

Inventory Service

Notification Service

A **Cloud Database** is a database that is hosted, managed, and accessed through a cloud platform instead of being installed on a local server.

Characteristics

- Fully managed by the cloud provider.
- Automatic scaling and high availability.
- Built-in backup and disaster recovery.
- Secure access with encryption and authentication.
- Accessible from anywhere via the Internet.

Examples

E-Commerce (Customer & Order)

Banking (Transaction Recs)

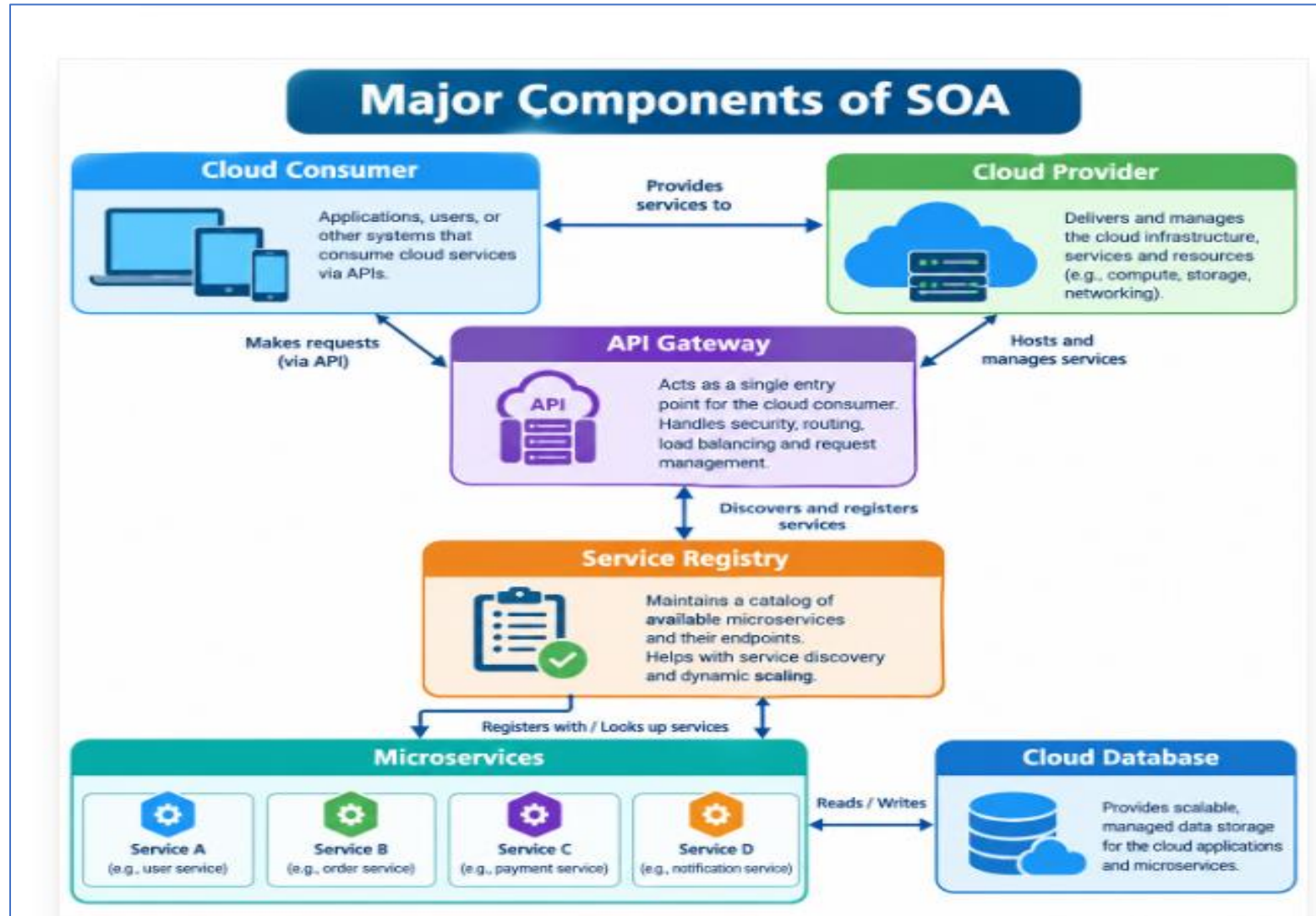
Healthcare (Patient Recs)

Social Media (User Profiles & Messages)

Online Learning (Student & Course Data)

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Cloud SOA: Components :



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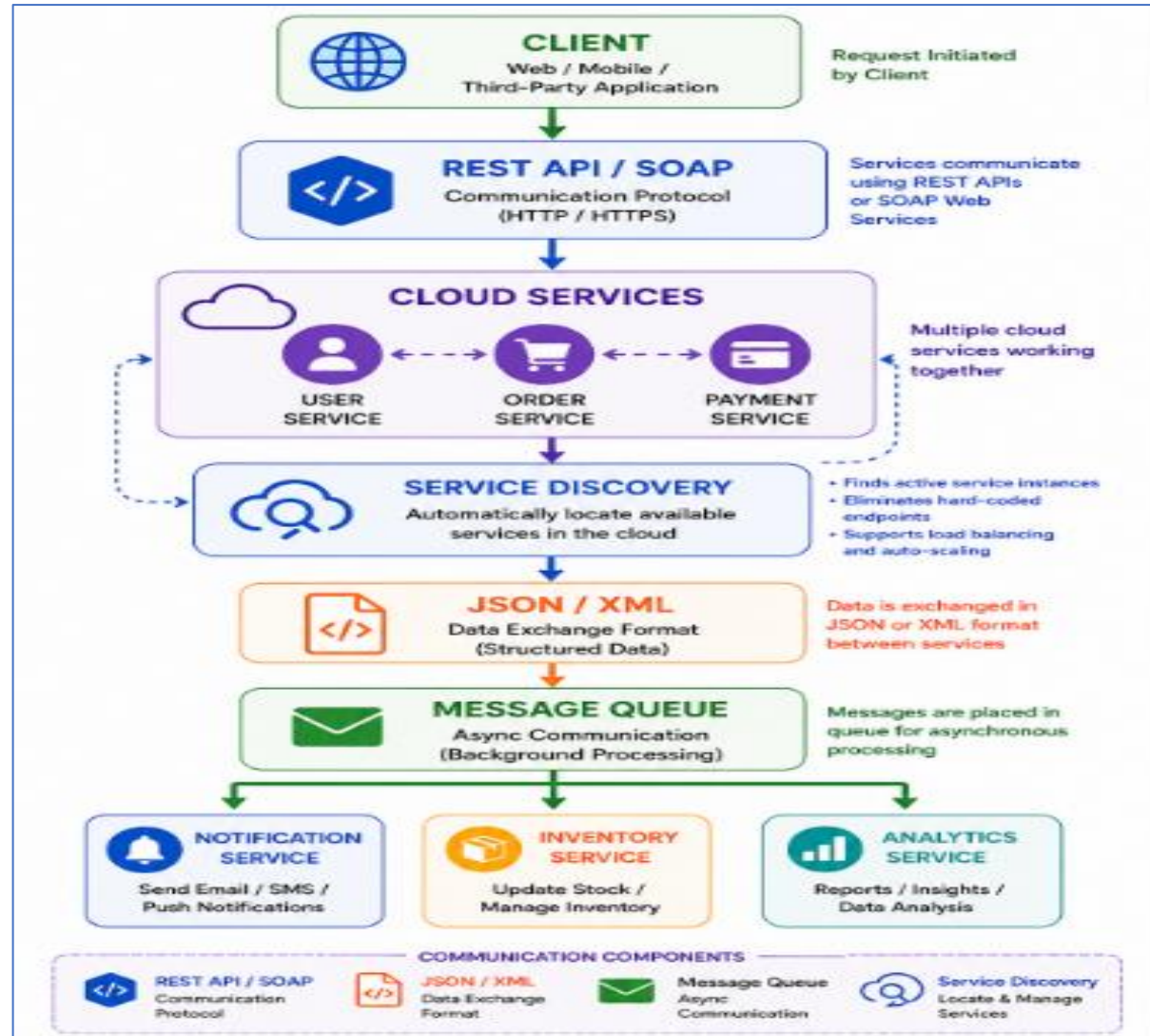
Cloud SOA: Communication Protocols

- SOA **Communication** is the process by which **independent cloud services communicate and exchange data** using standardized protocols and interfaces like.
- REST APIs (HTTP/HTTPS)
- SOAP Web Services
- XML & JSON Data Exchange
- Message Queues (Async Communication)
- Service Discovery

Component	Purpose
REST API	Real-time communication
SOAP	Secure enterprise communication
JSON/XML	Data exchange format
Message Queue	Asynchronous communication
Service Discovery	Locate available cloud services automatically

CLOUD TECHNOLOGIES

Cloud SOA: Communication Workflow



- SOA infrastructure provides the cloud resources required to deploy, execute, monitor, and manage services.

Major Components

1. Compute Services
2. Kubernetes
3. Networking Infrastructure
4. Storage & IAM

1. Compute Services

- Compute services provide the processing power needed to run applications, websites, and software. They include virtual machines, servers, and other resources that execute workloads.
- **Example:** AWS EC2/GCP/AZURE allows users to create virtual servers to host a website or run an application.

2. Kubernetes

- Kubernetes is an open-source platform used to **deploy, manage, and scale containerized applications**. It automatically manages containers, handles failures, and distributes applications across multiple servers.

***Containerized** means an application is packaged together with everything it needs to run—such as its code, libraries, dependencies, and settings—inside a **container**.*

- **Example:** A company can use Kubernetes to run a web application in several containers and automatically increase the number of containers when traffic increases.

3. Networking infrastructure

- **Networking infrastructure** connects computers, servers, applications, and users so that they can communicate and exchange data securely. It includes components such as routers, switches, firewalls, load balancers, and virtual networks.
- **Example:** A cloud application may use a Virtual Private Cloud (VPC), firewall rules, and a load balancer to securely connect users to application servers.

4. Storage & IAM

- **Storage** provides a place to save and retrieve data such as files, databases, backups, and application information, while **IAM (Identity and Access Management)** controls who can access those resources and what actions they are allowed to perform.
- **Example:** AWS S3 can store application files and backups, while AWS IAM can give an administrator permission to read and manage those files but allow a normal user to only view them.

Session Summary

- ✓ Define Cloud Service-Oriented Architecture (SOA).
- ✓ Explain the need for SOA in cloud computing.
- ✓ SOA Components
- ✓ SOA Communication
- ✓ SOA Infrastructure

Review Questions

1. Explain SOA with an example.
2. Describe SOA Components.
3. Differentiate between SOA Communication and Infrastructure.

1. What does Service-Oriented Architecture (SOA) mean? A. A method of storing data on a single server B. A programming language used for cloud applications C. An architecture in which applications are developed as a collection of independent, reusable services D. A hardware virtualization technology	2. Which of the following is responsible for providing a service to a service consumer in SOA? A. Service Registry B. Service Provider C. Hypervisor D. Database Server	3. Which component of SOA infrastructure helps different services communicate and integrate with each other? A. Compiler B. Operating System C. Enterprise Service Bus (ESB) D. Web Browser
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